SO(k)-Invariant Gauge Classification and the $\ell^1(E;\ell^2)$ Obstruction Geometry

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Abstract

Within the framework of cohomological obstruction theory, scalar ℓ^1 appears under replica-extensive, edge-decomposable observer assumptions. For vector-valued edge defects governed by a declared continuous rotational symmetry, maximal bounded-linear functional symmetry may be replaced by continuous rotational (SO(k)) invariant frames.

Under these explicit structural bounds, the scalar local gauge is replaced by the Euclidean local gauge, giving the mixed-norm obstruction geometry $J_{2,1}(\omega) = \sum_e c_e \|\omega_e\|_2$. The unweighted form $\sum_e \|\omega_e\|_2$ requires edge exchange symmetry or normalization.

This paper establishes the functional boundaries of this regime. **(1)** We formalize the primal $J_{2,1}$ obstruction problem for SO(k)-invariant local gauges over vector stalks, combined with standard norm axioms. **(2)** We prove the mixed-norm obstruction duality, demonstrating that the dual witnesses are Euclidean-unit-bounded, divergence-free vector circulations ($\|\varphi_e\|_2 \leq 1$). **(3)** We construct a finite triangle example separating coordinate-separable scalar observers from SO(k)-invariant vector observers.

This paper establishes the explicit two-part structural boundary for SO(k)-invariant vector-valued obstruction diagnostics: SO(k) symmetry selects the local ℓ^2 Euclidean gauge independently, and conditional Replica Extensivity (decomposability + disjoint additivity) selects the outer ℓ^1 edge sum. This bridges scalar metric constraints and Euclidean vector-valued obstruction geometries without conflating local symmetry boundaries with outer aggregate summation logic.

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1 Introduction and Structural Placement

The classification of diagnostic geometries over finite graphs forms the basis of categorical obstruction analysis. Previous classification theorems show that when an observer's operational domain demands edge-decomposable aggregation and maximal bounded-linear compatibility on scalar stalks, the resulting diagnostic is a weighted ℓ^1 coboundary norm, becoming uniform only under edge-uniformity assumptions.

However, moving from scalar signals to Euclidean geometries requires stating the structural constraints governing vector variables. Evaluating a 2D vector defect $\vec{\delta} = (\delta_x, \delta_y)$ via $|\delta_x| + |\delta_y|$ treats the two coordinates as independently observed scalar channels.

Conversely, treating $\vec{\delta}$ as a single vector under continuous rotation selects the local stalk gauge $\|\vec{\delta}\|_2$, up to scale.

The aggregate measuring these local bounds independently across edges is then the mixed $\ell^1(E;\ell^2)$ measure. This work formalizes the geometric separation between coordinate-separable scalar error components and vector-valued Euclidean defects.

2 General M-Compatible Gauge Classification

Evaluating obstruction geometries over finite carriers requires mapping local measurement constraints onto an admissible transformation monoid.

2.1 Definition

Let V be a finite-dimensional normed vector space and let

$$M \subset \mathrm{End}(V) \tag{1}$$

be an admissible transformation monoid representing local observer symmetries.

A faithful gauge $j : V \to \mathbb{R}_{\geq 0}$ is **M-compatible** if:

1. $j(x) = 0 \iff x = 0$
2. $j(\lambda x) = |\lambda| j(x)$
3. $j(x + y) \leq j(x) + j(y)$
4. $j(Tx) \leq \|T\| j(x)$ for all $T \in M$

Equivalently, j is a norm whose unit ball is compatible with the action of M.

Theorem 2.1 (General M-Compatible Gauge Classification)**.** *Let V be finite-dimensional and $M \subset \mathrm{End}(V)$ an admissible transformation monoid. There is a one-to-one correspondence between M-compatible faithful gauges j and closed, convex, balanced, absorbing unit bodies $B \subset V$ satisfying*

$$T(B) \subset \|T\| B \quad \textit{for all } T \in M. \tag{2}$$

The correspondence is given by the Minkowski functional $j_B(x) = \inf\{\lambda > 0 : x \in \lambda B\}$. Consequently, classifying M-compatible obstruction theories is equivalent to classifying M-contractive symmetric convex bodies. Increasing M shrinks the admissible body class. The maximal case $M = \mathrm{End}_b(V)$ collapses the body class to scalar multiples of the native Banach unit ball, recovering a unique local norm (up to scale); ℓ^1 arises only after imposing replica-extensive aggregation across edges.

2.2 Corollaries

- **Corollary 1 (Order Reversal):** If $M_1 \subset M_2$, then $\mathrm{Gauge}(V, M_2) \subset \mathrm{Gauge}(V, M_1)$. More symmetry means fewer admissible gauges.
- **Corollary 2 (Maximal Symmetry):** If $M = \mathrm{End}_b(V)$, then $\mathrm{Gauge}(V, M) = \{c\|\cdot\| : c > 0\}$. This recovers Paper 001.
- **Corollary 3 (Group Symmetry):** If $M = G$ is a group of isometries, then $j(gx) = j(x)$ for all $g \in G$, and the M-compatible gauges are exactly the G-invariant norms. Equivalently, B_j is a G-invariant symmetric convex body. If G acts transitively on the native unit sphere, then positive homogeneity and the triangle inequality isolate the local gauge up to scale as $j(x) = c\|x\|_2$.
- **Corollary 4 (Compact Group Construction):** If G is compact and q is any norm, then $j_G(x) := \int_G q(gx)dg$ is a G-invariant norm. Thus, compact symmetry classes generally admit vast continuous families of gauges unless the group acts transitively enough to collapse them.

- **Corollary 5 (Orbit Convex Hull Description):** For a group G, the minimal G-invariant gauge generated by a seed set $S \subset V$ has unit ball $B = \operatorname{conv}\left(\bigcup_{g \in G} gS\right)$, resolving constructively to the associated M-compatible gauge via its Minkowski functional.

Warning: There is generally **no finite list** of M-compatible gauges. The classification maps $M \to$ family of invariant/contractive convex bodies. Only restrictive maximal monoids collapse the hierarchy to a single local gauge class.

3 Formal Framework: Carrier and $\mathrm{SO}(k)$ Geometry

Let $(P, \leq)$ be a finite poset equipped with covering relations defining directed topological bonds E.

3.1 The Vector-Valued Carrier Stalks

Each vertex $v \in V$ carries a vector parameterization $s(v) \in \mathbb{R}^k$. For an edge $e = (a \to b)$, the declared edge comparison produces an offset defect:

$$\delta_e(s) = r_{b,a}(s(b)) - s(a) \tag{3}$$

where $\delta_e \in \mathbb{R}^k$.

3.2 The Core Theorem

Theorem 3.1. *Let P be a finite directed graph with edge defects in $\mathbb{R}^k$, $k \geq 2$. Suppose:*

1. *each local gauge j_e is a faithful convex gauge;*
2. *j_e is $\mathrm{SO}(k)$-invariant;*
3. *the global diagnostic is edge-decomposable;*
4. *the global diagnostic is additive over disjoint replicas;*
5. *edges are exchange-equivalent and normalized, or else edge weights c_e are retained.*

Then, under edge normalization,

$$J_{2,1}(\omega) = \sum_{e \in E} \|\omega_e\|_2. \tag{4}$$

Without edge normalization, the conclusion is the weighted mixed norm

$$J_{2,1}(\omega) = \sum_{e \in E} c_e \|\omega_e\|_2. \tag{5}$$

Proof. $\mathrm{SO}(k)$-invariance implies $j_e(x) = \phi_e(\|x\|_2)$ because $\mathrm{SO}(k)$ acts transitively on Euclidean spheres for $k \geq 2$. Positive homogeneity gives $\phi_e(r) = c_e r$. Hence $j_e(x) = c_e \|x\|_2$. Replica extensivity (edge decomposability and disjoint additivity) gives the outer sum. Edge symmetry and normalization give $c_e = 1$. □

Thus, $\mathrm{SO}(k)$, together with the norm/gauge axioms, selects the local ℓ^2 factor, while Replica Extensivity selects the outer ℓ^1 sum separately. Signed-permutation symmetry alone would not select ℓ^2; it admits many unconditional norms. We set $c_e = 1$ only under the stated edge-normalization hypothesis.

4 Primal Quotient Obstruction and Analytical Duality

When a system allows local adjustment fields $\alpha \in C^0(P, \mathbb{R}^k)$ as declared repairs, the residual class determines the **Quotient Obstruction** $\Phi_{2,1}$.

Definition 4.1 (Primal Form)**.**

$$\Phi_{2,1}([\omega]) = \inf_{\alpha} \sum_{e} \|\omega_e - (d_0\alpha)_e\|_2 \tag{6}$$

Theorem 4.2 (Mixed Vector Obstruction Duality)**.** *Let P be a finite directed graph with edge defects in $\mathbb{R}^k$. The primal quotient obstruction satisfies:*

$$\Phi_{2,1}([\omega]) = \max_{\varphi} \sum_{e} \langle \varphi_e, \omega_e \rangle \tag{7}$$

Subject to explicit structural bounds:

1. *Gauge Invariance / Divergence-Free Constraint: $d_0^*\varphi = 0$*
2. *Euclidean Unit-Sphere Witness Bounds: $\|\varphi_e\|_2 \leq 1$ for all edges $e \in E$.*

Proof. Write the primal problem as

$$\inf_{\alpha} \sum_{e} \|\omega_e - (d_0\alpha)_e\|_2. \tag{8}$$

The convex conjugate of $\|\cdot\|_2$ is the indicator function of the Euclidean unit ball. Introducing dual variables φ_e for each edge therefore imposes the pointwise bounds $\|\varphi_e\|_2 \leq 1$. Minimization over α imposes the stationarity constraint $d_0^*\varphi = 0$, i.e. the dual circulation is divergence-free. Since all spaces are finite-dimensional and the primal is feasible, Fenchel–Rockafellar duality gives equality of the primal infimum and dual supremum. Thus

$$\Phi_{2,1}([\omega]) = \max_{\substack{d_0^*\varphi=0 \\ \|\varphi_e\|_2 \leq 1}} \sum_{e} \langle \varphi_e, \omega_e \rangle. \tag{9}$$

□

5 Testable Triangle Prediction (The Falsifiable Fork)

To isolate the observer-level distinction, we identify a concrete finite cycle defect.

Consider a minimal closed triangle network $V = \{0, 1, 2\}$ with directed edges $E = \{(0, 1), (1, 2), (2, 0)\}$. Let the defect signal over the edges be:

- $\omega_{01} = (1, 0)$
- $\omega_{12} = (0, 1)$
- $\omega_{20} = (0, 0)$

The total cycle circulation evaluates to:

$$c = \omega_{01} + \omega_{12} + \omega_{20} = (1, 1) \tag{10}$$

For a single directed cycle with identity restrictions, the quotient class is determined by the total circulation $c = \sum_{e \in C} \omega_e$. Under the induced cycle quotient norm, this class has magnitude:

$$\Phi([\omega]) = j(c). \tag{11}$$

5.1 Prediction A — Scalar-Coordinate ℓ^1 ($J_{1,1}$)

If the topological observer separates variables (e.g. tracking independent coordinate channels): $J_{1,1}(\omega) = \sum_e(|\omega_{e,x}| + |\omega_{e,y}|)$ yields a quotient obstruction of:

$$\Phi_{1,1} = \|c\|_1 = |1| + |1| = 2 \tag{12}$$

5.2 Prediction B — Vector Mixed Norm ($J_{2,1}$)

If the topological observer operates under continuous SO(k) rotational invariance: $J_{2,1}(\omega) = \sum_e \|\omega_e\|_2$ yields a quotient obstruction of:

$$\Phi_{2,1} = \|c\|_2 = \sqrt{1^2 + 1^2} = \sqrt{2} \approx 1.414 \tag{13}$$

5.3 The Experimental Consequence

The two observers assign different quotient norms to the same cycle class.

A component-counting anomaly detector (separable coordinates, $J_{1,1}$) assesses the failure burden as **2**. A structural strain sensor bounding continuous Euclidean distances (SO(k), $J_{2,1}$) assesses the structural boundary as $\sqrt{2}$.

This distinguishes observer classes at the level of local gauge choice; it does not by itself test replica-extensive aggregation or dual structure.

6 Closure Invariant Bound Extension

Under topological evaluation, the 2-cell failure metric inherits the combined geometry. Since $d_1 d_0 = 0$, we have:

$$d_1(h + d_0\alpha) = d_1 h \tag{14}$$

so bounding the dual map norm yields:

$$\|d_1 h\|_{2,1} \leq \|d_1\|_{(2,1\to 2,1)} \|h + d_0\alpha\|_{2,1} \tag{15}$$

Taking the infimum over all α yields the quotient bound:

$$\Gamma_2([h]) = \sum_f \|(d_1 h)_f\|_2 \leq \|d_1\|_{(2,1\to 2,1)} \Phi_{2,1}([h]) \tag{16}$$

This guarantees the $\Phi\Gamma R_{cl}$ flow guardrails apply natively onto Euclidean rotational fields.

7 Conclusion

This work expands the obstruction topology by branching the theory into vector regimes. By combining continuous SO(k) rotational symmetry with standard homogeneous norm axioms, we isolate $\ell^1(E; \ell^2)$ as the geometry for observers whose local gauge is Euclidean and whose edge aggregation is replica-extensive. If edge exchange symmetry fails, the correct geometry is weighted $\ell^1(E; \ell^2)$. This preserves the analytical duality core while keeping the measurement geometry observer-relative.

7.1 Connection to the General Finite Obstruction Framework

The $\ell^1(E;\ell^2)$ geometry derived here integrates into the general finite obstruction calculus [1] as a specific gauge selection within the broader M-compatible family. Three connections are significant:

Observer passivity. In the general framework, observers are repair-compatible quotient maps satisfying $\Phi_{\mathcal{O}}(\mathcal{O}(q)) \leq \Phi(q)$ (Theorem O1 in [1]). When Φ is computed using the $\ell^1(E;\ell^2)$ geometry, the observer passivity bound constrains how SO(k)-measured defects can be processed by downstream observer maps. No observer quotient can create obstruction absent from the SO(k)-gauged core.

Contextual congruence. The lift trigger in the general framework is context-visible non-congruence: a witness $w = (x, y, c)$ with $p(x) = p(y)$ but $p(\rho(c)x) \neq p(\rho(c)y)$. The quotient obstruction $\Phi_{2,1}$ computed by the mixed norm determines *whether* the residual is nonzero; the context witness determines *whether a lift is licensed.* The testable triangle prediction (§5) should be read as an observer-class separation, not by itself as a context witness. It shows that separable-coordinate and SO(k)-invariant observers assign different quotient magnitudes to the same cycle class. A genuine context witness would require an additional declared context action $\rho(c)$ and a pair x, y with $p(x) = p(y)$ but $p(\rho(c)x) \neq p(\rho(c)y)$.

Sheaf gluing. The presheaf coboundary $\delta_e(s) = r_{b,a}(s(b)) - s(a)$ over vector stalks is exactly the local-to-global defect appearing in the sheaf gluing obstruction (Theorem L3 in [1]). When the presheaf stalks carry $\mathbb{R}^k$-valued data with SO(k) symmetry, the gluing obstruction is measured in the $\ell^1(E;\ell^2)$ geometry, and the dual witnesses are Euclidean-unit-bounded divergence-free vector circulations (§4.2).

These connections do not alter the core theorem. They provide the categorical governance within which the SO(k)-selected geometry operates.

References

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